

Quantum Field Theory — Lecture I

Dirac Equation & Relativistic Spinors

$$(i \cdot \gamma^\mu \cdot \partial_\mu - m) \cdot \psi = 0$$

$$E^2 = p^2 \cdot c^2 + m^2 \cdot c^4$$

$$\gamma^\mu \cdot \gamma^\nu + \gamma^\nu \cdot \gamma^\mu = 2 \cdot \eta^{\mu\nu}$$

Derivations & Physical Insights:

1. Relativistic invariance requires first-order time derivative.
2. Four-component Dirac spinor represents particle and antiparticle states with spin $\pm 1/2$.
3. Plane wave solutions: $\psi(x) = u(p) \cdot e^{(-i \cdot p \cdot x)}$.



(Feynman Diagram & Loop Vertex)

Gauge Invariance & QED Lagrangian

U(1) Gauge Principle & Minimal Coupling

$$\mathcal{L} = \bar{\psi} * (i * \gamma^\mu * D_\mu - m) * \psi - (1/4) * F_{\mu\nu} * F^{\mu\nu}$$

$$D_\mu = \partial_\mu + i * e * A_\mu$$

$$F_{\mu\nu} = \partial_\mu * A_\nu - \partial_\nu * A_\mu$$

Derivations & Physical Insights:

Local phase rotation: $\psi(x) \rightarrow \exp(i * \alpha(x)) * \psi(x)$

Gauge field transformation: $A_\mu(x) \rightarrow A_\mu(x) - (1/e) * \partial_\mu * \alpha(x)$

Ward-Takahashi identities enforce exact charge conservation.



[Feynman Diagram & Loop Vertex]

Feynman Path Integral Formulation

Functional Methods & Generating Functional

$$Z[J] = \text{Integral } D[\phi] * \exp(i * \text{Integral } d^4x (L + J * \phi))$$

$$\langle 0 | T\{\phi(x_1) \dots \phi(x_n)\} | 0 \rangle = (-i)^n * (\delta^n Z / \delta J(x_1) \dots \delta J(x_n)) | J=0$$

$$W[J] = -i * \ln(Z[J]) \quad (\text{Connected diagrams})$$

Derivations & Physical Insights:

Transition amplitude as sum over all possible field configurations.

Gaussian integration yields free propagator $\Delta_F(x - y)$.

Perturbative expansion generates standard Feynman diagrams and loop vertex corrections.



(Feynman Diagram & Loop Vertex)

Renormalization & Loop Corrections

One-Loop Vacuum Polarization & Beta Function

$$\Pi^{\mu\nu}(q) = (q^2 * \eta^{\mu\nu} - q^\mu * q^\nu) * \Pi(q^2)$$

$$\alpha(\mu^2) = \alpha(\mu_0^2) / (1 - (2*\alpha / 3*\pi) * \ln(\mu/\mu_0))$$

$$\beta(g) = \mu * (d g / d \mu) = - (11*C_A - 4*n_f*T_F) / (12*\pi) * g^3$$

Derivations & Physical Insights:

Dimensional regularization: $d = 4 - 2 * \epsilon$.

Running coupling constant: Screening vs Anti-screening in non-Abelian Yang-Mills theory.

Asymptotic freedom at high energy scales (Nobel Prize in Physics).



[Feynman Diagram & Loop Vertex]